From Person-Specific Dynamics to Population Inference: A Meta-Analytic Structural Equation Modeling Framework for Multilevel Dynamic Models

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Intensive longitudinal designs—such as ecological momentary assessment (EMA) and other experience-sampling, daily diary, ambulatory assessment, and measurement-burst protocols—have made it increasingly feasible to collect dense within-person data in everyday contexts (Bolger & Laurenceau, 2013; Bolger et al., 2003; Csikszentmihalyi & Larson, 1987; Nesselroade, 1991; Shiffman et al., 2008; Sliwinski, 2008; Trull & Ebner-Priemer, 2013). These designs make it increasingly feasible to study psychological processes as dynamic systems, where within-person trajectories evolve over time and individuals differ meaningfully in the parameters that govern those trajectories (Asparouhov et al., 2018; Bringmann et al., 2013, 2016; Fisher et al., 2011; Hamaker et al., 2018; Rowland & Wenzel, 2020). This shift has reinforced a long-standing methodological point: inferences about within-person dynamics cannot be taken for granted from between-person variation, and scientific questions about regulation, coupling, and stability often require models that respect the distinction between idiographic (person-specific) and nomothetic (population-level) structure (Curran & Bauer, 2011; Fisher et al., 2018; Gates et al., 2023; Hamaker et al., 2005; Molenaar, 2004; Wang et al., 2026). At the same time, applied researchers rarely want only a collection of person-specific estimates; they also want principled summaries of what is typical, how much people differ, how dynamic features co-vary across people, and how these dynamic "signatures" relate to covariates and distal outcomes (Beltz et al., 2016; Li et al., 2022). Yet existing workflows often force a trade-off between (a) fully pooled multilevel or Bayesian models that can be computationally demanding and shrink person-specific dynamics and (b) "fit-many-then-summarize" approaches that are fast but typically ignore first-stage uncertainty and the multivariate dependence among dynamic parameters—limiting system-level inference and rapid model iteration (Moeyaert et al., 2016; Van den Noortgate & Onghena, 2008).

To meet these applied goals, we introduce MetaVAR, a meta-analytic structural equation modeling (MASEM; Cheung, 2008, 2013, 2015) framework for vector autoregressive (VAR; Hamilton, 1994; Lütkepohl, 2005) systems. MetaVAR[1] is a two-stage approach that (a) estimates person-

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[1]We use the label MetaVAR because, in this manuscript, the Stage 1 idiographic model is a VAR fit separately to each individual. Importantly, the proposed two-stage framework is not tied to VAR dynamics. Stage 2 requires only that each person contribute (a) a vector of estimated model features, $\mathbf{y}_i$, and (b) an accompanying sampling covariance matrix, $\mathbf{V}_i^2$, so that first-stage uncertainty is propagated into the population synthesis. Thus, MetaVAR may be viewed as a special case of a more general meta-analytic dynamical-systems template in which person-specific pa-

specific VAR dynamics and (b) performs *system-level* synthesis of the resulting parameter vectors using multivariate random-effects MASEM with full propagation of first-stage uncertainty. By carrying forward each individual's full first-stage sampling covariance, MetaVAR yields population summaries of typical dynamics, heterogeneity, and cross-parameter covariation. When desired, the same synthesis stage can be extended to relate dynamic signatures to covariates and distal outcomes while retaining Stage 1 error propagation.

Against this backdrop, the key methodological challenge is to draw population-level conclusions about within-person dynamics without sacrificing inferential validity or practical feasibility. Current workflows tend to fall at two ends of a spectrum: fully pooled hierarchical models that can be computationally demanding and induce shrinkage in person-specific dynamics, and fully unpooled fit-many approaches that scale well but often treat first-stage estimates as error-free. In the next section, we review these dominant strategies and argue that both leave a persistent gap: a principled two-stage workflow that (a) performs system-level synthesis with full propagation of Stage 1 uncertainty and (b) supports theory-driven constraints, with optional extensions for covariate moderation and distal outcomes. This gap motivates MetaVAR, which we evaluate in a Monte Carlo simulation, illustrate with an empirical example, and then discuss in terms of practical guidance, limitations, and future extensions.

Two Dominant Strategies and a Persistent Gap

Two broad strategies dominate current practice for drawing population-level conclusions from intensive longitudinal multivariate time series. The first is *fully pooled* (one-step) modeling—including multilevel and Bayesian multilevel formulations such as dynamic structural equation modeling (DSEM) and Bayesian multilevel VAR (mlVAR)—in which within-person dynamics and between-person variation are estimated simultaneously within a single hierarchical model (Asparouhov et al., 2018; Hamaker et al., 2018; Li et al., 2022). This approach is attractive in the behavioral sciences because the per-person series length is often limited, making it beneficial to "borrow strength" across individuals while still allowing random effects in dynamic parameters (Li et al., 2022). At the same time, the resulting estimation problem can become computationally demanding as the number of variables and random effects grows, and practical implementations often require simplifying assumptions about the random-effects structure or otherwise face feasibility and convergence constraints in high-dimensional settings (Asparouhov et al., 2018; Epskamp et al., 2018; Li et al., 2022). Moreover, because hierarchical estimators partially pool information, person-specific dynamics are shrunken toward the population mean—a feature that can improve stability and out-of-sample performance (Gelman & Hill, 2006). At the same time, shrinkage can make genuine between-person heterogeneity less visible in the resulting individual-level estimates (and in downstream summaries built from them), potentially obscuring extreme dynamic regimes (Efron & Morris, 1975; Gelman & Hill, 2006).

At the other extreme are *fully unpooled* workflows that fit an idiographic model per person and then summarize the resulting coefficients informally (e.g., by averaging). This "fit-many-then-summarize" approach is modular and naturally parallelizable, but naïve summaries ignore that individuals' estimates differ in precision as a function of series length, missingness, and measurement quality, and simple averaging fails to weight estimates by their uncertainty (Hedges & Olkin, 1985; Pastor & Lazowski, 2017). In intensive time-series settings, dependence and model misspecification (e.g., unmodeled trends) can also induce artifactual residual autocorrelation and larger residuals, further affecting precision and inference if not modeled (Baek et al., 2023; Shadish et al., 2008). In multivariate dynamic systems, a second limitation is that piecemeal estimation (e.g., collections of univariate models) need not place all parameters in a single joint model, making it difficult to retain the within-person sampling covariance among dynamic parameters that is required for principled system-level synthesis and inference (Epskamp et al., 2018; Moeyaert et al., 2016). Accordingly, valid two-stage synthesis requires carrying forward each individual's first-stage sampling variance (and, in multivariate settings, the full sampling covariance) into the second stage rather than treating person-level estimates as error-free (Van den Noortgate & Onghena, 2008).

A natural middle ground is *two-stage* inference: estimate person-specific dynamic models first, then integrate those results to make population statements about within-person processes. However, two-stage workflows only deliver valid population inference when the second stage correctly accounts for first-stage estimation error and, in multivariate settings, the dependence among estimated parameters. The present work develops a two-stage framework that fills this gap for dynamic systems by treating the synthesis stage as a multivariate meta-analytic structural equation model.

Related Two-Stage Literatures: Idiographic Meta-Analysis and Error-Corrected Pooling

MetaVAR builds on, and is intended to sit alongside, two closely related two-stage literatures. First, Lee and Gates (2023) advocate a *two-stage idiographic meta-analytic* perspective for observational time series, emphasizing the con-

rameters (or derived functionals thereof) from a wide class of dynamical models—including state-space, continuous-time, and nonlinear formulations—are synthesized using the same multivariate mixed-effects SEM machinery.

ceptual importance of estimating within-person models idiographically and then using random-effects meta-analysis to describe population distributions of within-person effects. This line foregrounds heterogeneity as a scientific target; MetaVAR adopts that stance and shares its two-stage spirit, but it broadens the target of synthesis. Whereas idiographic meta-analytic work often aggregates parameters marginally using a diagonal weight matrix, MetaVAR synthesizes the joint distribution of a unit's dynamic parameters by carrying forward the full Stage-1 sampling covariance. Doing so preserves dependence among within-person effects, supports coherent inference under constraints, and enables valid uncertainty propagation to functions of the dynamics (e.g., stability indices, impulse responses, and contemporaneous/temporal network summaries).

Second, Zhang et al. (2025) develop an *error-corrected two-stage* approach for integrating individual-level dynamic factor analysis (DFA) parameters, demonstrating that ignoring first-stage estimation error can bias second-stage random-effects estimation and showing how to correct this efficiently. MetaVAR generalizes the same error-propagation logic to VAR systems and leverages an SEM-based synthesis stage to support *system-level* inference: joint hypotheses, cross-parameter constraints, and multivariate moderation/outcome models defined on the full parameter vector. Taken together, these strands motivate a two-stage workflow that is both conceptually idiographic and inferentially principled; MetaVAR operationalizes this combination for VAR models by casting the second stage as multivariate meta-analytic structural equation modeling.

MetaVAR: Dynamic MASEM for System-Level Pooling With Full Error Propagation

MetaVAR is a two-stage framework that treats multilevel dynamic modeling as a meta-analytic problem in which *persons play the role of studies*. In Stage 1, each individual $i = 1, \dots, N$ is fit with an idiographic dynamic model (e.g., VAR, latent-variable VAR, or state-space model), yielding a vector of estimated dynamic parameters and an accompanying sampling covariance matrix. In Stage 2, these person-level parameter vectors are synthesized using a multivariate mixed-effects meta-analytic model expressed as a structural equation model. This design yields *system-level pooling with full error propagation*: MetaVAR preserves the full within-person sampling covariance among estimated parameters from Stage 1 and simultaneously estimates a between-person random-effects covariance structure for the entire parameter system. The perspective connects directly to multivariate random-effects meta-analysis (Berkey et al., 1998; van Houwelingen et al., 1993, 2002) and MASEM (Cheung, 2008, 2013, 2015) and directly targets questions central to intensive longitudinal research: What is the typical within-person dynamic system? How heterogeneous is it? Which dynamic features co-occur across people? Which person-level covariates predict dynamic regimes? How do dynamics relate to clinically meaningful distal outcomes?

Stage 1 Outputs as Multivariate Effect Sizes

Let $\mathbf{y}_i \in \mathbb{R}^p$ denote the vector of person-i parameter estimates returned by the Stage 1 model, treated as a multivariate set of meta-analytic effect sizes, and let $\mathbf{V}_i^2 \in \mathbb{R}^{p\times p}$ denote the corresponding Stage 1 sampling covariance matrix. The contents of $\mathbf{y}_i$ are deliberately flexible: depending on the Stage 1 model, $\mathbf{y}_i$ may include autoregressive and cross-lag coefficients, innovation variances and covariances, set point (mean) parameters, measurement-error variances, or other person-specific system features. The only requirement for MetaVAR is that Stage 1 yields $\mathbf{V}_i^2$ (or a good approximation), so that Stage 2 can propagate Stage 1 uncertainty rather than treating the person-level estimates as error-free.

In this manuscript, Stage 1 fits a person-specific first-order VAR model. Let $\mathbf{s}_{i,t} \in \mathbb{R}^k$ denote the observed multivariate time series for person i at occasion $t = 1, \dots, T$. The dynamic model follows a centered VAR model (Asparouhov & Muthén, 2018; Hamaker & Grasman, 2015),

$$\mathbf{s}_{i,t} = \boldsymbol{\mu}_i + \mathbf{A}_i\left(\mathbf{s}_{i,t-1} - \boldsymbol{\mu}_i\right) + \boldsymbol{\zeta}_{i,t}, \quad \boldsymbol{\zeta}_{i,t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Xi}_i), \tag{1}$$

where $\boldsymbol{\mu}_i$ is the person-specific set point (i.e., the equilibrium or long-run mean level of the process), $\mathbf{A}_i$ collects autoregressive and cross-lagged effects that govern the evolution of deviations from this set point, and $\boldsymbol{\Xi}_i$ is the innovation covariance. Under stability/stationarity (e.g., all eigenvalues of $\mathbf{A}_i$ lie inside the unit circle) and mean-zero innovations, the process fluctuates around $\boldsymbol{\mu}_i$ such that $\mathbb{E}(\mathbf{s}_{i,t}) = \boldsymbol{\mu}_i$.

To connect Stage 1 to Stage 2, we stack the person-specific parameters into a single vector $\boldsymbol{\theta}_i \in \mathbb{R}^p$ and take $\mathbf{y}_i = \hat{\boldsymbol{\theta}}_i$. A convenient choice for the present model is

$$\boldsymbol{\theta}_i = (\boldsymbol{\mu}_i', \mathrm{vec}(\mathbf{A}_i)', \mathrm{vech}(\boldsymbol{\Xi}_i)')', \tag{2}$$

where $(\cdot)'$ represents matrix transpose, vec$(\cdot)$ stacks a matrix by columns and vech$(\cdot)$ stacks the lower triangle (including the diagonal) of a symmetric matrix. The associated Stage 1 sampling covariance matrix $\mathbf{V}_i^2 = \widehat{\mathbb{V}}\left[\hat{\boldsymbol{\theta}}_i\right]$ can be obtained from the observed information matrix, a sandwich estimator, or a parametric bootstrap, and is passed to Stage 2 to propagate first-stage estimation uncertainty. If some components are fixed or omitted in a given application, they are simply excluded from $\boldsymbol{\theta}_i$ and the dimension p is adjusted accordingly.

Stage 2 as Multivariate Mixed-Effects Meta-Analysis in SEM Form

A general mixed-effects meta-analytic SEM can be written as (Cheung, 2008, 2013)

$$\mathbf{y}_i = \boldsymbol{\alpha}_i + \boldsymbol{\beta}\mathbf{y}_i + \boldsymbol{\Gamma}\mathbf{x}_i + \boldsymbol{\varepsilon}_i, \quad \boldsymbol{\varepsilon}_i \sim \mathcal{N}\left(\mathbf{0}, \mathbf{V}_i^2\right), \tag{3}$$

with person-specific "true" effect vectors

$$\boldsymbol{\alpha}_i = \boldsymbol{\alpha} + \boldsymbol{\upsilon}_i, \quad \boldsymbol{\upsilon}_i \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\tau}^2\right). \tag{4}$$

Here, $\boldsymbol{\alpha}$ is the grand-mean vector of the dynamic-parameter system, $\boldsymbol{\tau}^2$ is the between-person heterogeneity covariance matrix, $\mathbf{x}_i$ collects person-level covariates (e.g., traits, clinical status, context), and $\boldsymbol{\beta}$ and $\boldsymbol{\Gamma}$ allow researchers to specify structured relations and meta-regressions (moderation) defined *on the parameter system* itself.

Assuming that $\mathbf{x}_i$ is non-stochastic and that $\mathbf{I} - \boldsymbol{\beta}$ is non-singular, the model can be written in reduced form (Bollen, 1989) as

$$\mathbf{y}_i = (\mathbf{I} - \boldsymbol{\beta})^{-1} (\boldsymbol{\alpha}_i + \boldsymbol{\Gamma}\mathbf{x}_i + \boldsymbol{\varepsilon}_i). \tag{5}$$

Therefore, conditional on $\mathbf{x}_i$, the marginal model implied mean vector and covariance matrix[2] are

$$\mathbb{E}[\mathbf{y}_i \mid \mathbf{x}_i] = (\mathbf{I} - \boldsymbol{\beta})^{-1} (\boldsymbol{\alpha} + \boldsymbol{\Gamma}\mathbf{x}_i), \tag{6}$$

$$\mathbb{V}[\mathbf{y}_i \mid \mathbf{x}_i] = (\mathbf{I} - \boldsymbol{\beta})^{-1} \left(\boldsymbol{\tau}^2 + \mathbf{V}_i^2\right) (\mathbf{I} - \boldsymbol{\beta})^{-1'}. \tag{7}$$

In the special case $\boldsymbol{\beta} = \mathbf{0}$, the model reduces to a multivariate mixed-effects meta-analysis (i.e., multivariate meta-regression when $\boldsymbol{\Gamma} \neq \mathbf{0}$), with $\mathbb{E}[\mathbf{y}_i \mid \mathbf{x}_i] = \boldsymbol{\alpha} + \boldsymbol{\Gamma}\mathbf{x}_i$ and $\mathbb{V}[\mathbf{y}_i \mid \mathbf{x}_i] = \boldsymbol{\tau}^2 + \mathbf{V}_i^2$. When no covariates are included and $\boldsymbol{\beta} = \mathbf{0}$, the model further reduces to a multivariate random-effects meta-analysis with $\mathbb{E}[\mathbf{y}_i] = \boldsymbol{\alpha}$ and $\mathbb{V}[\mathbf{y}_i] = \boldsymbol{\tau}^2 + \mathbf{V}_i^2$. Thus, the model decomposes observed dispersion in $\mathbf{y}_i$ into (a) within-person estimation uncertainty carried by $\mathbf{V}_i^2$ and (b) genuine between-person heterogeneity carried by $\boldsymbol{\tau}^2$ (Pastor & Lazowski, 2017).

This system-level viewpoint is the key departure from edge-by-edge pooling. MetaVAR treats the full parameter vector as the unit of synthesis, which (a) retains the entire within-person sampling covariance among dynamic parameters and (b) estimates a coherent between-person covariance structure for the dynamic system. As a descriptive summary, heterogeneity can be reported using multivariate generalizations of $\mathbf{I}^2$ (Higgins & Thompson, 2002; Jackson et al., 2012; Viechtbauer, 2007), for example

$$\mathbf{I}^2 = p^{-1} \left\{ \operatorname{tr} \left[\boldsymbol{\tau}^2 \left(\boldsymbol{\tau}^2 + \bar{\mathbf{V}}^2 \right)^{-1} \right] \right\}. \tag{8}$$

Here, $\operatorname{tr}(\cdot)$ denotes the matrix trace (the sum of diagonal elements), and $\bar{\mathbf{V}}^2$ is the harmonic (precision-weighted) mean of the Stage 1 sampling covariance matrices,

$$\bar{\mathbf{V}}^2 = \left[N^{-1} \sum_{i=1}^{N} \left(\mathbf{V}_i^2 \right)^{-1} \right]^{-1}, \tag{9}$$

which gives greater weight to individuals with more precise Stage 1 estimates (smaller sampling variances). Values near zero indicate that most observed dispersion in person-level parameters reflects estimation uncertainty, whereas values near one indicate substantial true heterogeneity in the underlying dynamic system.

Model-Level Hypotheses and Constraints

Because Stage 2 is an SEM defined on the full parameter vector, MetaVAR is naturally aligned with *model-level* hypotheses—that is, hypotheses that constrain or compare *multiple* dynamic parameters jointly rather than treating each coefficient as an isolated "edge." This matters for applied research because many substantive predictions about regulation are inherently multivariate: for example, whether cross-lag effects are asymmetric ($A_{1,2} \neq A_{2,1}$), whether a subset of couplings is negligible relative to others (block-zero constraints), whether inertia parameters are comparable across constructs (equality constraints), or whether a theoretically motivated contrast among parameters differs from zero (e.g., a "self–other" coupling contrast).

In MetaVAR, these hypotheses are expressed directly through standard SEM constraint machinery applied to $\boldsymbol{\alpha}$, $\boldsymbol{\tau}^2$, $\boldsymbol{\beta}$, and $\boldsymbol{\Gamma}$. This yields several practical benefits. First, inference is *system-aware*: constraints are evaluated while retaining the within-person sampling covariance $\mathbf{V}_i^2$ and the between-person heterogeneity structure $\boldsymbol{\tau}^2$, so uncertainty is not understated by treating estimated parameters as independent or error-free. Second, constraints can be placed not only on individual coefficients but also on *functions* of the dynamic system that are often the actual scientific target (e.g., stability-related summaries, coupling differences, or composite indices that pool information across parameters). Third, competing theory-level models can be compared transparently using standard SEM tools (e.g., likelihood-based model comparison for nested specifications or information criteria for non-nested structures), enabling researchers to formalize "which dynamic story fits best" at the level of the full parameter system.

In this manuscript, we focus on the core two-stage synthesis: person-specific dynamic parameters are first estimated idiographically, then synthesized using multivariate random-effects meta-analysis (cast as meta-analytic SEM) with full propagation of each individual's Stage 1 sampling covari-

[2]In SEM software that supports *definition variables*, elements of the known, person-specific sampling covariance matrix $\mathbf{V}_i^2$ can be supplied as case-level data so that the model-implied covariance is recomputed for each individual (Mehta & Neale, 2005). This capability is central in meta-analytic applications because $\mathbf{V}_i^2$ is treated as a known input (and can be fixed as such), whereas common mixed-effects implementations (e.g., `lmer`) do not directly support fixing a case-specific sampling covariance matrix in the likelihood (Declercq et al., 2020). In `OpenMx`, this is implemented by evaluating each individual's multivariate normal log-likelihood using that individual's model-implied mean vector and covariance matrix, which may depend on definition variables (FIML; Arbuckle, 1996; Enders, 2010; Neale et al., 2015). The `metaSEM` package leverages this capability via `OpenMx` (Cheung, 2015); the `metaDyn` implementation of MetaVAR uses the same definition-variable strategy to incorporate $\mathbf{V}_i^2$ and propagate Stage 1 uncertainty through Stage 2.

ance matrix. The same SEM template can be extended to incorporate covariates, distal outcomes, and other second-stage structures, but we treat these as future directions and return to them in the General Discussion.

Evaluation and Empirical Illustration

The remainder of the paper is organized to both evaluate the framework's operating characteristics and demonstrate its applied utility. After presenting the Stage 2 multivariate mixed-effects SEM formulation and the distal-outcome extension, we evaluate MetaVAR in a Monte Carlo simulation that examines performance across design factors and data-generating scenarios (varying N and T). The simulation focuses on recovery of population-average dynamics and heterogeneity, as well as uncertainty quantification when person-specific Stage 1 sampling covariance matrices are propagated into Stage 2. We then illustrate the end-to-end workflow in an empirical example using intensive longitudinal affect data and intake assessments, showing how MetaVAR yields interpretable system-level summaries and supports principled inference about distal outcomes from person-specific dynamic signatures. We conclude with a general discussion that synthesizes the methodological contributions, offers practical guidance for applied researchers, and highlights limitations and directions for future extensions.

Monte Carlo Simulation Study

We conducted a Monte Carlo simulation study to evaluate the operating characteristics of MetaVAR under a controlled bivariate affect-dynamics data-generating process (DGP), and to compare its performance against two commonly used alternatives: a Bayesian multilevel VAR (BMLVAR) approach and a naïve "fit-many-then-summarize" approach. The primary goal was to assess how well each method recovers (a) population-average dynamic parameters and (b) between-person heterogeneity in those parameters across combinations of sample size (N) and per-person time series length (T). We also evaluated uncertainty quantification, with particular attention to whether accounting for first-stage estimation uncertainty improves second-stage inference.

Data-Generating Process

Data were generated from the same centered first-order VAR model used in Stage 1 of MetaVAR:

$$\mathbf{s}_{i,t} = \boldsymbol{\mu}_i + \mathbf{A}_i\left(\mathbf{s}_{i,t-1} - \boldsymbol{\mu}_i\right) + \boldsymbol{\zeta}_{i,t}, \quad \boldsymbol{\zeta}_{i,t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Xi}),$$

for person $i = 1, \ldots, N$ and occasion $t = 1, \ldots, T$. The person-specific set points were generated as

$$\boldsymbol{\mu}_i \sim \mathcal{N}\left(\begin{pmatrix} 2.87 \\ 2.04 \end{pmatrix}, \begin{pmatrix} 1.20 & 0.46 \\ 0.46 & 1.10 \end{pmatrix}\right),$$

and the person-specific transition matrices were generated by drawing

$$\operatorname{vec}(\mathbf{A}_i) \sim \mathcal{N}\left(\begin{pmatrix} 0.280 \\ -0.035 \\ -0.048 \\ 0.260 \end{pmatrix}, \begin{pmatrix} 0.017 & 0.005 & 0.001 & 0.008 \\ 0.005 & 0.008 & 0.001 & 0.006 \\ 0.001 & 0.001 & 0.001 & 0.002 \\ 0.008 & 0.006 & 0.002 & 0.026 \end{pmatrix}\right).$$

To ensure a stationary VAR process for each individual, generated transition matrices were retained only if all eigenvalues had moduli less than 1. Thus, the simulation targets the setting in which person-specific dynamics are heterogeneous but dynamically stable. The process-noise covariance matrix was held constant across persons and occasions,

$$\boldsymbol{\Xi} = \begin{pmatrix} 1.30 & 0.57 \\ 0.57 & 1.56 \end{pmatrix},$$

so that the simulation isolates method performance for recovering heterogeneity in the set points and lagged dynamics. Fixing $\boldsymbol{\Xi}$ also simplifies interpretation of differences across methods because any between-person variation in estimated innovation parameters arises from finite-sample estimation error rather than true heterogeneity in the DGP.

The parameter values were informed by prior empirical work on affective dynamics (e.g., Bringmann et al., 2013), but were not copied exactly. In particular, we parameterized the person-specific *mean* (set point) $\boldsymbol{\mu}_i$ rather than the intercept vector. This choice is substantively convenient because $\boldsymbol{\mu}_i$ directly represents each person's typical equilibrium level in the centered VAR parameterization, whereas the intercept in the uncentered form is a transformed quantity (i.e., a function of both the mean and transition matrix). The average transition matrix implies moderate positive autoregressive persistence (inertia) for both variables and small negative cross-lagged effects, representing a stable bivariate regulatory system with weak reciprocal inhibitory coupling on average. The nonzero covariance among elements of vec($\mathbf{A}_i$) further induces realistic heterogeneity in dynamic signatures across individuals.

Design Factors

We crossed three sample sizes and three numbers of measurement occasions:

$$N \in \{50, 100, 200\}, \qquad T \in \{50, 100, 200\}.$$

This yielded a 3×3 factorial design with nine conditions. These values were chosen to represent common ranges in intensive longitudinal research, spanning small-to-moderate numbers of participants and short-to-moderate person-level series lengths.

For each condition, we generated multiple Monte Carlo replications ($R = 1,000$). In each replication, person-specific time series were generated under the DGP above, analyzed by each competing method, and summarized using the performance metrics described below.

Methods Compared

We compared three approaches that represent the main inferential strategies discussed in the Introduction: (a) a two-stage method with explicit propagation of first-stage uncertainty (MetaVAR), (b) a one-step hierarchical estimator (Bayesian multilevel VAR; BMLVAR), and (c) a two-stage baseline that does not propagate first-stage uncertainty (naïve fit-many-then-summarize). The comparison was designed to isolate the consequences of (i) one-step versus two-step estimation and (ii) whether uncertainty in person-specific Stage 1 estimates is carried forward into population-level inference.

Across all methods, the target of inference was the same: population-average dynamic parameters (fixed effects) and between-person heterogeneity in those parameters (random effects), as defined by the simulation DGP. To facilitate comparability, all methods were fit to the same simulated datasets under each (N, T) condition, and summaries were mapped to a common parameterization whenever possible (i.e., set points/means, VAR transition parameters, and between-person variability in those parameters). Differences across methods therefore reflect estimation strategy and inferential assumptions rather than differences in data input.

MetaVAR (proposed)

MetaVAR was fit as a two-stage procedure. In Stage 1, a person-specific VAR(1) model was fit separately for each individual to obtain (a) a vector of estimated dynamic parameters and (b) the corresponding estimated sampling covariance matrix for that parameter vector. The Stage 1 outputs were then passed to Stage 2, where the person-level parameter vectors were synthesized using a multivariate random-effects meta-analytic model (implemented in SEM form). This Stage 2 model treats the person-specific sampling covariance matrices as known inputs, thereby propagating first-stage estimation uncertainty into the population-level likelihood.

The key inferential feature of MetaVAR is that it operates on the *full parameter vector* for each person rather than pooling parameters one at a time. Consequently, MetaVAR retains the within-person sampling covariance among estimated dynamic parameters in Stage 2 and estimates a coherent between-person random-effects covariance structure across the parameter system. In the simulation, we evaluated MetaVAR using both model-based (normal-theory) standard errors and robust (sandwich) standard errors at Stage 2 to assess sensitivity of inference to potential misspecification and finite-sample behavior.

Bayesian Multilevel VAR (BMLVAR)

As a one-step comparator, we fit a Bayesian multilevel VAR model to the full stacked multilevel time-series data. Under this approach, person-specific dynamics and population-level parameters are estimated jointly within a single hierarchical model. In contrast to MetaVAR, which separates idiographic estimation (Stage 1) from population synthesis (Stage 2), BMLVAR performs simultaneous estimation and naturally induces partial pooling (shrinkage) of person-specific parameters toward the population distribution.

BMLVAR was included because it represents a widely used and substantively appealing framework for multilevel intensive longitudinal data, especially when per-person time series are short and borrowing strength across individuals can stabilize estimation. At the same time, joint estimation and shrinkage make BMLVAR an informative benchmark against MetaVAR, which is intentionally designed to preserve a modular two-stage workflow while still supporting principled population inference. For BMLVAR, uncertainty was summarized using posterior standard deviations of the target parameters, and inferential sensitivity was summarized using posterior credible-interval exclusion of zero (see Performance Metrics).

Naïve "Fit-Many-Then-Summarize"

As a baseline two-stage approach, person-specific VAR(1) models were fit separately and then summarized across individuals without propagating first-stage estimation uncertainty in the second stage. Population-average parameters were obtained by direct aggregation of person-specific estimates (e.g., arithmetic means), and uncertainty for the population-average estimates was quantified using the standard error of the mean across persons.

This approach was included because it reflects a common practical workflow in idiographic time-series research: fit models person-by-person, extract parameters of substantive interest, and then summarize those parameters descriptively across individuals. Its advantages are modularity, transparency, and computational simplicity. However, because the second-stage summaries do not explicitly account for differences in precision of the person-specific estimates (e.g., due to finite T or estimation instability), the naïve approach provides a useful baseline for quantifying the inferential gains from MetaVAR's error-propagating synthesis stage.

Comparative expectations. The three methods differ in ways that are directly relevant to the goals of the simulation. Relative to the naïve approach, MetaVAR is expected to improve population-level inference by weighting and synthesizing person-specific estimates while carrying forward their sampling uncertainty. Relative to BMLVAR, MetaVAR trades one-step joint estimation for a modular two-stage workflow that preserves idiographic estimation and supports system-level meta-analytic synthesis. The simulation therefore evaluates not only point-estimate recovery but also the inferential consequences of these distinct estimation strategies.

Performance Metrics

We evaluated performance separately for (a) population-average (fixed-effect) parameters and (b) between-person heterogeneity (random-effect) parameters.

Point-Estimate Performance

For each target parameter θ (e.g., an element of the population mean vector or heterogeneity covariance matrix), we computed Monte Carlo relative bias and RMSE across replications:

$$\text{Relative Bias}(\hat{\theta}) = \frac{\bar{\hat{\theta}} - \theta}{\theta}, \tag{10}$$

$$\text{RMSE}(\hat{\theta}) = \sqrt{\frac{1}{R}\sum_{r=1}^{R}\left(\hat{\theta}^{(r)} - \theta\right)^2}, \tag{11}$$

where $\bar{\hat{\theta}} = R^{-1}\sum_{r=1}^{R}\hat{\theta}^{(r)}$.

Because relative bias can be unstable for parameters near zero (notably the cross-lagged effects in the present DGP), we additionally recommend reporting absolute bias (or signed bias in the original metric) for those parameters.

Uncertainty Quantification

We evaluated standard errors (or posterior standard deviations) for population-average parameters and, when applicable, heterogeneity parameters. For each method, we summarized:

1. the mean estimated standard error across replications,
2. the empirical standard deviation of the point estimates across replications,
3. the ratio of mean estimated SE to empirical SD (values near 1 indicate calibrated uncertainty estimates).

For MetaVAR, we report both normal-theory and robust (sandwich) standard errors. For the naïve approach, we report the standard error of the mean across individuals. For the Bayesian multilevel model, we report posterior standard deviations.

Hypothesis-Testing Performance (Empirical Power)

Because the present DGP does not include exact-zero target parameters, we evaluated inferential sensitivity using empirical power (i.e., empirical detection rates) for parameters with nonzero true values rather than empirical Type I error.

For a given target parameter θ and nominal level α (e.g., $\alpha = .05$), empirical power was defined as

$$\widehat{\text{Power}}(\theta) = \frac{1}{R}\sum_{r=1}^{R}\mathbb{I}(\text{reject } H_0 : \theta = 0 \text{ in replication } r), \tag{12}$$

where $\mathbb{I}(\cdot)$ is the indicator function.

For MetaVAR and the naïve approach, rejection was based on two-sided Wald-type tests (equivalently, whether the nominal 95% confidence interval excluded zero). For MetaVAR, we report results using both normal-theory and robust (sandwich) standard errors. For the Bayesian multilevel model, we report the empirical detection rate based on whether the nominal 95% posterior credible interval excluded zero. Although this Bayesian criterion is not a frequentist hypothesis test, it provides a practically comparable summary of inferential sensitivity across methods.

Because some true effects in the DGP are small in magnitude (especially the cross-lagged effects), empirical power is interpreted jointly with bias, RMSE, and uncertainty calibration.

Simulation Targets

The simulation primarily focused on recovery of:

1. the population-average set points and transition parameters (fixed effects),
2. the between-person heterogeneity in these parameters (random-effects variances/covariances),
3. the accuracy of uncertainty estimates under each method.

This design allows us to test the core claim motivating MetaVAR: that propagating person-specific Stage 1 sampling covariance into a multivariate second-stage synthesis improves population-level inference relative to naïve aggregation, while retaining the computational and conceptual advantages of a modular two-stage workflow.

Monte Carlo Simulation Results

Monte Carlo Simulation Discussion

Empirical Illustration: Meta-Analysis of Daily Affect Dynamics

We illustrate the proposed MetaVAR workflow using intensive longitudinal data from the Affective Dynamics and Individual Differences study (ADID; Emotions & Dynamic Systems Laboratory, 2010), which was designed to examine how emotion dynamics unfold in everyday life and how these dynamics differ across individuals. The ADID EMA protocol sampled participants five times per day for approximately

30 days (four daytime prompts plus an evening report), yielding up to 150 measurement occasions per participant in the raw data. ADID has been published in part across multiple substantive and methodological papers, including work on regime-switching dynamics, measurement/trait–state structure, network and VAR-based modeling, outlier-robust state-space modeling, and missing-data methods (Chow & Zhang, 2013; Gates et al., 2023; Guo et al., 2012; Hutton & Chow, 2014; Li et al., 2022, 2024; Oh et al., 2025; Ou et al., 2023; Park et al., 2020; You et al., 2019).

This design directly matches the core goal of MetaVAR: estimating person-specific dynamic "signatures" in a first stage and then synthesizing them at the population level in a second stage while propagating each individual's Stage 1 sampling uncertainty. In the present illustration, we focus on within-person dynamics between negative affect (NA) and positive affect (PA). Specifically, we meta-analyze the full person-specific parameter system consisting of (a) affective set points, (b) temporal dynamics encoded by the VAR(1) transition matrix—autoregressive (AR) inertia and cross-regressive (CR) coupling—and (c) the process-noise (innovation) covariance matrix, which captures within-occasion co-fluctuation of unexpected changes after accounting for lagged dynamics.

Substantively, we focus on NA and PA because emotion-dynamics theories treat affect as a self-regulating system characterized by (at least) a typical level (often conceptualized as a "set point") and temporal persistence from one occasion to the next (often conceptualized as affective "inertia"), with potential cross-lagged coupling between NA and PA reflecting carryover and regulatory spillover (Houben et al., 2015; Kuppens, Allen, & Sheeber, 2010; Kuppens, Oravecz, & Tuerlinckx, 2010). These features are of substantive interest because they capture how quickly individuals return to (or remain near) their typical affective state after perturbations. MetaVAR treats heterogeneity in these features—including heterogeneity in the innovation covariance structure—as a scientific target, providing population-average inferences while quantifying reliable between-person variation and co-variation across elements of the dynamic system.

Measures

Affect Variables (Stage 1 Dynamics)

Momentary affect in ADID was assessed using adjective ratings drawn from established affect instruments, including items from the Positive Affect and Negative Affect Schedule (PANAS; Watson et al., 1988) and circumplex-content items used to broaden coverage of activation levels (Russell, 1980). Participants rated how often they experienced each affect descriptor since the previous assessment on an ordinal Likert-type scale, as implemented in the ADID daily diary surveys. Using the recorded date–time stamps, we rounded responses to the nearest hour and then constructed an hourly, regularly spaced time series by inserting missing values for hours with no report. Following prior ADID applications, we construct per-occasion NA and PA composites by averaging the available negative-affect and positive-affect items, respectively. To reduce the influence of slow drift over the month-long protocol and to better align the series with the stationary VAR(1) assumption, we apply a within-person linear detrending step to each affect composite prior to model fitting. In the VAR(1) model, person-specific set points represent typical affect levels, the transition matrix encodes temporal persistence and cross-lagged coupling, and the innovation covariance matrix characterizes same-occasion co-fluctuation of unexpected deviations after accounting for lagged dynamics.

Empirical Example Results

Stage 1 estimated person-specific NA–PA dynamics for each participant, and Stage 2 synthesized these individual "dynamic signatures" to obtain (a) population-average dynamics, (b) between-person heterogeneity in these dynamics, and (c) systematic co-variation among elements of the parameter system. Below we summarize the main results in this order.

Stage 1

We fit the person-specific VAR(1) dynamics model in Equation 1 separately for each participant, estimating all dynamic parameters as individual-specific (i.e., person-specific set points, autoregressive coefficients, cross-regressive coefficients, and innovation covariance matrix). Models were fit for $N = 217$ individuals. Of these, 210 individual models converged successfully and were retained for the Stage 2 synthesis; 7 individual models failed to converge and were excluded from subsequent analyses. Thus, Stage 2 results are based on $N = 210$ individuals with converged Stage 1 estimates.

Stage 2: Multivariate Random-Effects Meta-Analysis of the Dynamic Parameter System

Stage 2 synthesized the person-specific VAR(1) parameter systems using a multivariate random-effects meta-analysis. The meta-analytic mean vector $\boldsymbol{\alpha}$ contained nine elements, ordered as: the two set points ($\mu_{\text{NA}}, \mu_{\text{PA}}$), the four VAR transition coefficients ($\beta_{11}, \beta_{21}, \beta_{12}, \beta_{22}$) corresponding to AR and cross-regressive (CR) effects, and the three unique elements of the process-noise covariance matrix $\boldsymbol{\Psi}$ for NA and PA ($\psi_{11}, \psi_{21}, \psi_{22}$).

Population-average dynamics (α). The population-average set points indicated higher typical PA than NA: $\mu_{\text{NA}} = 1.4308$ (SE = 0.0255), 95% CI [1.3809, 1.4808], and $\mu_{\text{PA}} = 2.5935$ (SE = 0.0331), 95% CI [2.5287, 2.6583]. The average temporal dynamics showed positive inertia in both affect dimensions, with $\beta_{11} = 0.3668$ (SE = 0.0303), 95% CI [0.3074, 0.4262] for NA and $\beta_{22} = 0.3158$ (SE = 0.0318), 95% CI [0.2534, 0.3781] for PA. Cross-regressive effects were negative on average: $\beta_{21} = -0.0669$ (SE = 0.0349), 95% CI [-0.1353, 0.0015] for NA→PA (marginal at $p = .055$) and $\beta_{12} = -0.0491$ (SE = 0.0196), 95% CI [-0.0875, -0.0107] for PA→NA ($p = .012$).

Stage 2 also estimated the population-average process-noise covariance matrix in the original metric:

$$\hat{\boldsymbol{\Psi}} = \begin{pmatrix} \psi_{11} & \psi_{21} \\ \psi_{21} & \psi_{22} \end{pmatrix} = \begin{pmatrix} 0.0331 & -0.0079 \\ -0.0079 & 0.0624 \end{pmatrix},$$

with $\psi_{11} = 0.0331$ (SE = 0.0019), 95% CI [0.0293, 0.0368], $\psi_{21} = -0.0079$ (SE = 0.0010), 95% CI [-0.0098, -0.0061], and $\psi_{22} = 0.0624$ (SE = 0.0030), 95% CI [0.0564, 0.0683]. Thus, after accounting for lagged dynamics, the innovation variances were larger for PA than NA, and the innovation covariance was negative, indicating that within-occasion shocks to NA and PA tended to move in opposite directions on average.

Between-person heterogeneity and co-variation ($\boldsymbol{\tau}^2$). The estimated random-effects covariance matrix $\boldsymbol{\tau}^2$ indicated substantial heterogeneity across the full parameter system. Diagonal elements (between-person variances) were: 0.1320 for μ_{NA} (SD $\approx$ 0.36), 0.2223 for μ_{PA} (SD $\approx$ 0.47), 0.1654 for β_{11} (SD $\approx$ 0.41), 0.1910 for β_{21} (SD $\approx$ 0.44), 0.0640 for β_{12} (SD $\approx$ 0.25), 0.1854 for β_{22} (SD $\approx$ 0.43), and (for the innovation covariance parameters) 0.0006 for ψ_{11} (SD $\approx$ 0.025), 0.0001 for ψ_{21} (SD $\approx$ 0.010), and 0.0016 for ψ_{22} (SD $\approx$ 0.040). Several off-diagonal elements were also statistically distinguishable from zero, indicating systematic co-variation among components of the dynamic signature across individuals. For example, NA inertia β_{11} covaried negatively with NA→PA coupling β_{21} (cov = −0.0456, $p = .004$), and PA inertia β_{22} covaried negatively with PA→NA coupling β_{12} (cov = −0.0246, $p = .008$). Co-variation was also present within the innovation-covariance parameters (e.g., ψ_{11} with ψ_{21} and ψ_{22}; all $p < .001$), consistent with meaningful between-person differences in the magnitude and co-fluctuation of within-occasion shocks.

Heterogeneity index (I^2). The I^2 indices were uniformly high for all nine elements of $\boldsymbol{\alpha}$, indicating that the observed dispersion in Stage 1 parameter estimates was overwhelmingly attributable to true between-person heterogeneity rather than within-person sampling variability. Specifically, I^2 ranged from 0.974 for β_{21} to essentially 1.000 for the set points and innovation-variance parameters (e.g., $I^2 = 1.000$ for μ_{NA} and ψ_{11}), with the remaining parameters also above 0.989. Taken together, the $\boldsymbol{\tau}^2$ and I^2 results support the core premise motivating MetaVAR: affective set points, temporal dynamics (AR/CR), and innovation covariance structure all vary reliably across individuals, making multivariate synthesis of the full parameter system scientifically informative.

Empirical Example Discussion

This empirical illustration demonstrates how MetaVAR can be used to (a) estimate idiographic affect-dynamics parameters from intensive longitudinal data and (b) synthesize these person-specific "dynamic signatures" into population-average inferences while quantifying heterogeneity (Emotions & Dynamic Systems Laboratory, 2010; Fisher et al., 2017). In ADID, the approach yielded coherent population-average dynamics, strong evidence of between-person heterogeneity in both set points and lagged dependencies, and interpretable co-variation among elements of the parameter system (Houben et al., 2015). In addition to the familiar *temporal* dependencies (AR and CR effects in the VAR transition matrix), MetaVAR also supports population inference on the *contemporaneous* layer implied by the process innovations—summarized here via partial correlations derived from the innovation precision matrix—which captures same-occasion conditional co-fluctuation after accounting for lagged dynamics.

Population-Average NA–PA Dynamics

At the population level, participants reported higher typical PA than NA, consistent with findings from intensive longitudinal studies of community samples in which average affective tone tends to be more positive than negative (Houben et al., 2015; Watson et al., 1988). Both NA and PA exhibited positive inertia, with AR(NA) slightly larger than AR(PA), suggesting that momentary elevations in NA tended to persist somewhat more strongly than momentary elevations in PA at the hourly scale used here (Houben et al., 2015; Kuppens, Oravecz, & Tuerlinckx, 2010). Cross-lagged coupling was negative in both directions: higher NA predicted subsequent decreases in PA, and higher PA predicted subsequent decreases in NA. Interpreted as a bivariate self-regulating system, these average cross-lagged effects are consistent with reciprocal inhibition/spillover processes that have been discussed in affect-dynamics theory (Kuppens, Allen, & Sheeber, 2010; Kuppens, Oravecz, & Tuerlinckx, 2010). Importantly, these are descriptive features at a specific time scale (hourly, after within-person detrending) and should be interpreted as average short-term dependencies rather than as mechanistic causal effects (Asparouhov & Muthén, 2018; Hamaker & Grasman, 2015).

Heterogeneity as a Substantive Finding, not a Nuisance

The random-effects variance components indicated substantial between-person variability across the dynamic signature, including both set points and lagged dynamics. The corresponding $\mathbf{I}^2$ values were uniformly high, implying that the observed dispersion in Stage 1 estimates overwhelmingly reflects true between-person differences rather than sampling variability of the person-specific estimates (Higgins & Thompson, 2002). Methodologically, the results motivate multivariate synthesis (rather than edge-by-edge pooling) because heterogeneity is present across parameters and the off-diagonal elements of $\boldsymbol{\tau}^2$ can reveal how deviations in set points, inertia, coupling, and innovation structure co-occur across individuals (Cheung, 2015). This heterogeneity can likewise be interpreted in a two-layer manner: individuals may differ in temporal persistence and delayed coupling (elements of the transition matrix) and/or in contemporaneous co-fluctuation of innovations (functionals of $\boldsymbol{\Xi}$, such as innovation partial correlations derived from $\boldsymbol{\Xi}^{-1}$). Framing heterogeneity this way helps separate differences in *carry-over* dynamics from differences in *within-occasion coupling* of unexpected affective shifts.

Practical and Methodological Considerations

Several features of the analysis highlight common trade-offs in applied dynamic modeling. First, a small fraction of Stage 1 models did not converge, underscoring the importance of diagnostics and robustness checks in idiographic estimation (e.g., sensitivity to starting values, stationarity constraints, alternative innovation structures, or mild regularization) (Gates et al., 2023; Ou et al., 2023). Second, we imposed regular spacing by rounding to the nearest hour and inserting missing values, and we applied within-person linear detrending to better align the data with a stationary VAR(1) approximation; both steps improve feasibility but may also alter the interpretation of parameters (e.g., detrending shifts focus from slower drifts to shorter-term fluctuations) (Guo et al., 2012; Li et al., 2022). Third, NA/PA composites were constructed from ordinal item ratings and treated as approximately continuous; future work could examine whether a latent measurement model or item-level modeling changes the estimated dynamic signature and its heterogeneity (Park et al., 2020; Watson et al., 1988).

Implications and Future Directions

Overall, the illustration supports the substantive value of treating affect dynamics as person-specific signatures that can be meaningfully compared and synthesized (Fisher et al., 2017; Houben et al., 2015). Several extensions follow naturally from the same Stage 2 SEM template but are not pursued in the present illustration, including (a) covariate moderation of the parameter system and (b) linking dynamic signatures to distal outcomes, both under full Stage 1 error propagation. Additional directions include allowing nonstationarity (e.g., regime switching or time-varying parameters) (Chow & Zhang, 2013; Hutton & Chow, 2014), extending Stage 1 to continuous-time formulations to better handle irregular sampling without discretization (Park et al., 2025), and targeting multivariate constraints or derived functionals (e.g., stability summaries, impulse-response metrics, or contemporaneous network summaries) as higher-level quantities for synthesis (Cheung, 2015).